

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL GCSE BIOLOGY

Paper 2

Tuesday 2 November 2021

07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a ruler with millimetre measurements
- a scientific calculator.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice

In all calculations, show clearly how you work out your answer.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	

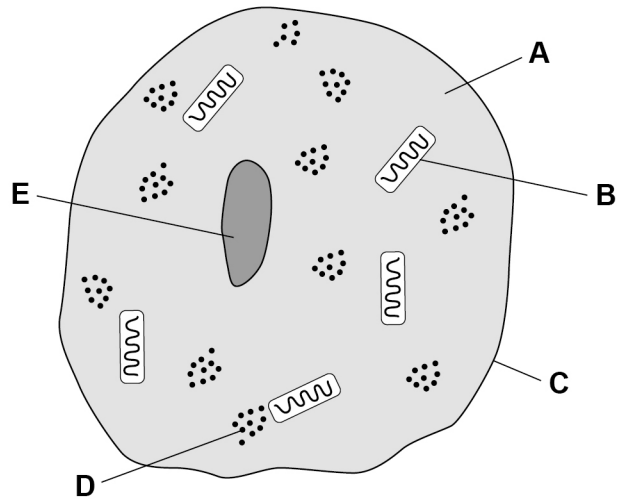


Answer **all** questions in the spaces provided.

0 1

Figure 1 shows an animal cell.

Figure 1



0 1 . 1

Name part A.

[1 mark]

A _____

0 1 . 2

Which part controls the movement of substances into and out of the cell?

[1 mark]

Tick (✓) **one** box.

B

☐

C

☐

D

☐

E

☐


0 1 . 3 Which part is where protein synthesis occurs?

[1 mark]

Tick (✓) **one** box.

B

☐

C

☐

D

☐

E

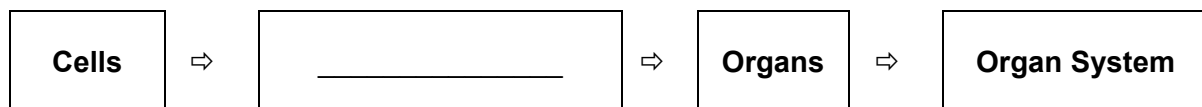
☐

0 1 . 4 **Figure 2** shows how cells are organised into an organ system in the human body.

Complete **Figure 2**.

[1 mark]

Figure 2



0 1 . 5 The human body has many organ systems.

Draw **one** line from each organ to the organ system it is part of.

[2 marks]

Organ

Brain

Heart

Stomach

Organ system

Circulatory system

Digestive system

Nervous system

Turn over ►

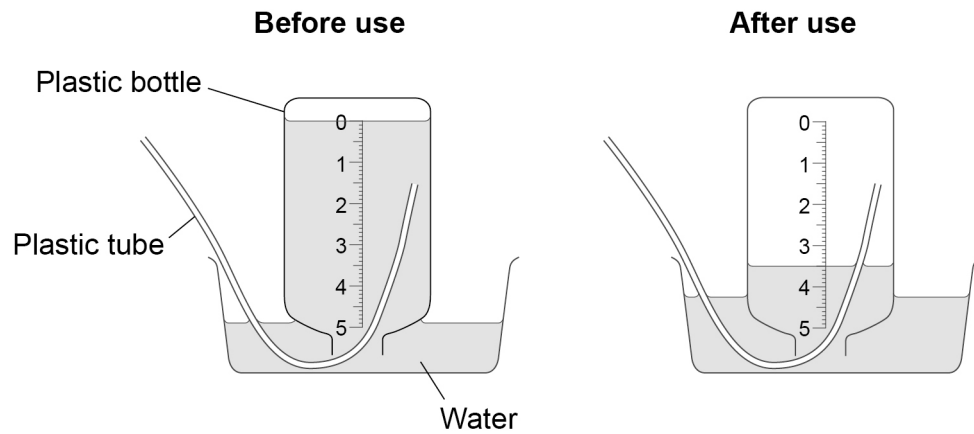


The breathing system takes air into and out of the lungs.

The maximum volume of air breathed out of the lungs in one breath can be measured.

Figure 3 shows the equipment which can be used.

Figure 3



0 1 . 6

Describe how you could use the equipment in **Figure 3** to measure the **maximum** volume of air breathed out in one breath.

[3 marks]



A student investigated the relationship between the height of a person and the volume of air they could breathe out in one breath.

This is the method used.

1. Choose 10 male students and divide into two groups of 5 students:
 - **Group 1** above 1.70 m in height
 - **Group 2** below 1.65 m in height.
2. Measure the volume of air breathed out by each student in one breath.
3. Calculate the mean volume of air breathed out for each group.

0 1 . 7

What was the independent variable for the investigation?

[1 mark]

Tick (✓) **one** box.

Height of the students

☐

Sex of the students

☐

Type of equipment used

☐

0 1 . 8

Suggest why the students in group 1 could breathe out a larger volume of air than the students in group 2.

[2 marks]

12

Turn over for the next question

Turn over ►



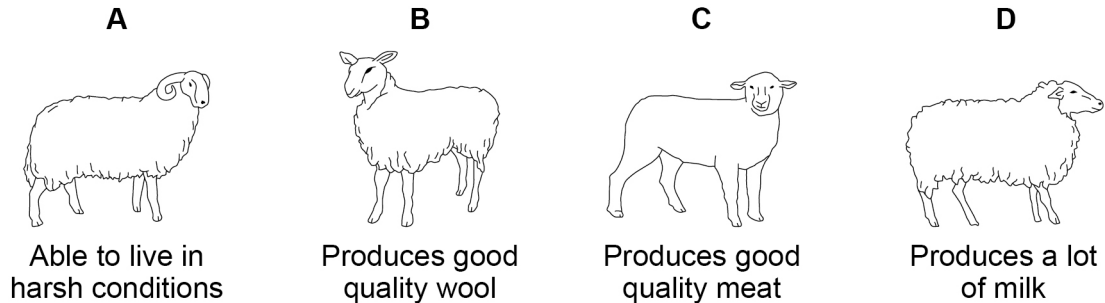
0 2

When sheep are bred the offspring have some characteristics of each parent.

0 2 . 1

Figure 4 shows the characteristics of four types of sheep.

Figure 4



A farmer wants sheep that can survive in very cold conditions and produce good quality meat.

Which **two** sheep should the farmer use to breed offspring with these characteristics?

Tick (✓) **one** box.

[1 mark]

Sheep **A** and sheep **C**

☐

Sheep **A** and sheep **D**

☐

Sheep **B** and sheep **C**

☐

Sheep **B** and sheep **D**

☐


The farmer wanted to breed sheep with longer wool.

This is the method used.

1. Choose five female sheep and five male sheep with the longest wool.
2. Allow to breed.
3. Wait for the offspring to reach maturity.
4. Choose offspring with the longest wool and allow to breed.
5. Repeat steps 3 and 4 for several generations.

0 2 . 2

Why is this method **not** an example of natural selection?

[1 mark]

0 2 . 3

After eight years the farmer had sheep with longer wool.

Suggest why this took a long time to achieve.

[1 mark]

Question 2 continues on the next page

Turn over ►



Table 1 shows the heritability value of some characteristics in sheep.

The heritability value is an estimate of how likely it is that a characteristic is due to inherited genes.

Table 1

Characteristic	Heritability value
Body mass at birth	0.15
Body mass of adult	0.40
Depth of body fat	0.23
Length of wool	0.30
Milk production	0.10

The higher the heritability value the more likely it is that the characteristic is due to inherited genes.

- A low heritability value is < 0.15
- A high heritability value is > 0.30

0 2 . 4

Name a characteristic with a low heritability value.

[1 mark]

0 2 . 5

Name a characteristic with a high heritability value.

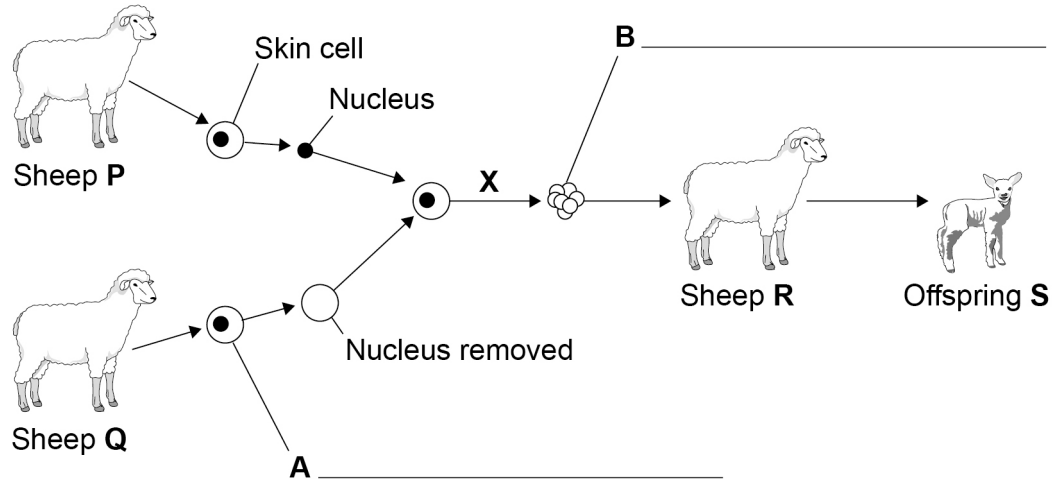
[1 mark]



Sheep can be cloned using a technique called adult cell cloning.

Figure 5 shows the main stages in the adult cell cloning of sheep.

Figure 5



0 2 . 6 Label **A** and **B** on **Figure 5**.

Choose answers from the box.

[2 marks]

cytoplasm

egg

embryo

gene

sperm

0 2 . 7 Why was the cell given an electric shock at **X**?

[1 mark]

Question 2 continues on the next page

Turn over ►



0 2 . 8 Offspring **S** is genetically identical to **one** of the sheep in **Figure 5**.

Which sheep, **P**, **Q**, or **R**, is offspring **S** genetically identical to?

Give the reason for your answer.

[2 marks]

Sheep _____

Reason _____

10



Turn over for the next question

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ANSWER IN THE SPACES PROVIDED**

Turn over ►



0 3

Cress is a small plant which is quick and easy to grow.

Figure 6 shows cress seeds and cress plants.

Figure 6

Cress seeds



Cress plants



A student investigated the effect of the space available on the growth of cress plants.

This is the method used.

1. Place cotton wool into a Petri dish and moisten with water.
2. Spread 0.06 g of cress seeds evenly onto the surface of the cotton wool.
3. Leave the dish in a room at 20 °C.
4. Keep the cotton wool moist by adding water when necessary.
5. After 14 days record the length of 20 plant stems.
6. Repeat steps 1 – 5 using different masses of seeds.

0 3 . 1

The mass of 20 cress seeds is 0.06 grams.

The student used this value to calculate the number of seeds put into each dish.

Calculate the number of seeds used if a mass of 0.39 grams of seeds was weighed.

[3 marks]

Number of seeds = _____



0 3 . 2 Suggest why the student did **not** count the number of seeds placed into each dish.

[1 mark]

0 3 . 3 The student measured the length of the plant stems.

Give one **other** measurement of plant growth the student could take.

[1 mark]

0 3 . 4 Give **two** factors that could affect the length of the plant stems.

Do **not** refer to availability of space in your answer.

[2 marks]

1

2

Question 3 continues on the next page

Turn over ►



Table 2 shows the results.

Table 2

Mass of seeds in dish in grams	Calculated number of seeds in the dish	Range of length of stems in mm	Mean length of stems in mm
0.06	20	37 to 54	46
0.15	50	56 to 65	60
0.30	100	35 to 53	44
0.60	200	38 to 52	40
0.90	300	33 to 48	40

0 3 . 5 What was the resolution of the balance the student used?

Tick (✓) **one** box.

[1 mark]

0.01 g

☐

0.1 g

☐

10 g

☐

100 g

☐


0 3 . 6

Complete **Table 3** to describe the effects of increasing the mass of seeds on the mean length of stems.

Use information from **Table 2** on page 14.

Choose answers from the box.

[2 marks]

decreases	increases	stays constant
-----------	-----------	----------------

Table 3

Mass of seeds in dish in grams	Effect on the mean length of stems
From 0.06 to 0.15	
From 0.15 to 0.60	
From 0.60 to 0.90	

0 3 . 7

What was the range for the length of stems recorded in the investigation?

Use information from **Table 2** on the opposite page.

[1 mark]

Range = from _____ mm to _____ mm

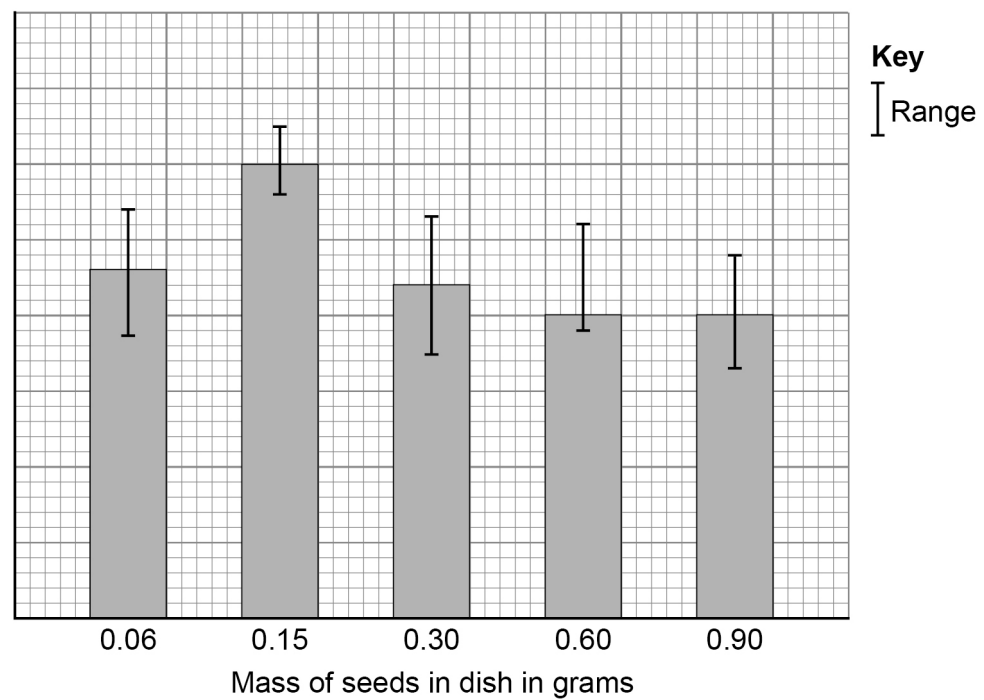
Question 3 continues on the next page

Turn over ►



Figure 7 shows the results from **Table 2**.

Figure 7



0 3 . 8

Complete **Figure 7**.

You should:

- label the y-axis
- add the scale for the y-axis.

[2 marks]

0 3 . 9

The range on **Figure 7** shows the spread of results about the mean value.

Which mass of seeds gave results that showed the greatest precision?

[1 mark]

Mass = _____ g

14



0 4

Microorganisms which cause disease are called pathogens.

0 4 . 1Name **one** type of pathogen.**[1 mark]**

0 4 . 2

Pathogens can make people feel ill.

Give **two** reasons why pathogens make people feel ill.**[2 marks]**1

2

0 4 . 3

White blood cells produce antibodies to help defend the human body against pathogens.

Give **two** other ways that white blood cells help to defend the human body against pathogens.Do **not** refer to antibodies in your answer**[2 marks]**1

2

Question 4 continues on the next page**Turn over ►**

One type of pathogen causes influenza in horses (horse flu).

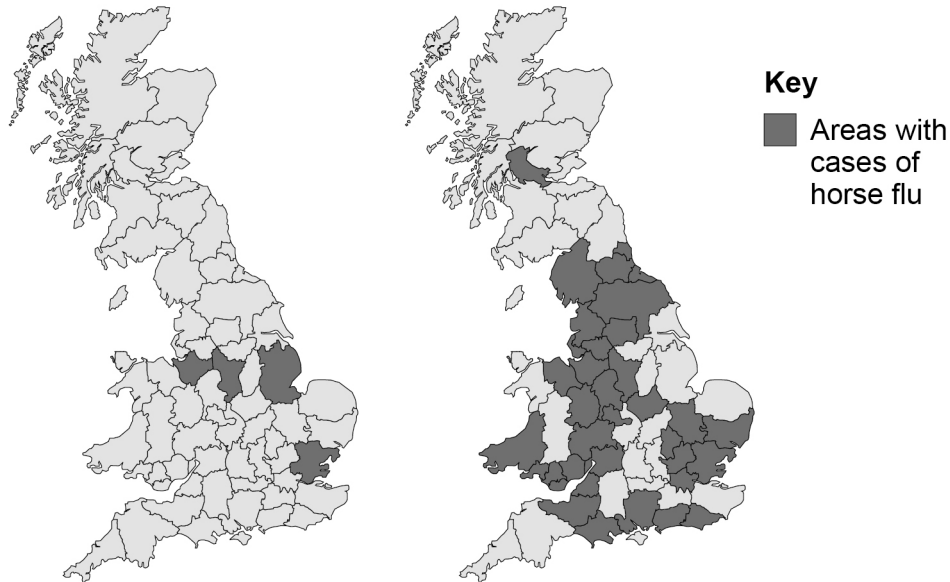
0 4 . 4

Figure 8 gives information about the cases of horse flu in parts of the UK in January 2019 and in June 2019.

Figure 8

5 cases in January

64 cases in June



Describe **two** changes in the cases of horse flu from January 2019 to June 2019.

[2 marks]

- 1 _____
- 2 _____



A vaccine can protect horses against horse flu.

0 4 . 5

In one year there was a total of 165 cases of horse flu.

17 horses had been vaccinated out of the 165 horses with horse flu.

Calculate the ratio of vaccinated to non-vaccinated horses.

Give your answer to 3 significant figures.

[3 marks]

Ratio of vaccinated horses: non-vaccinated horses = _____ : 1
(to 3 significant figures)

Question 4 continues on the next page

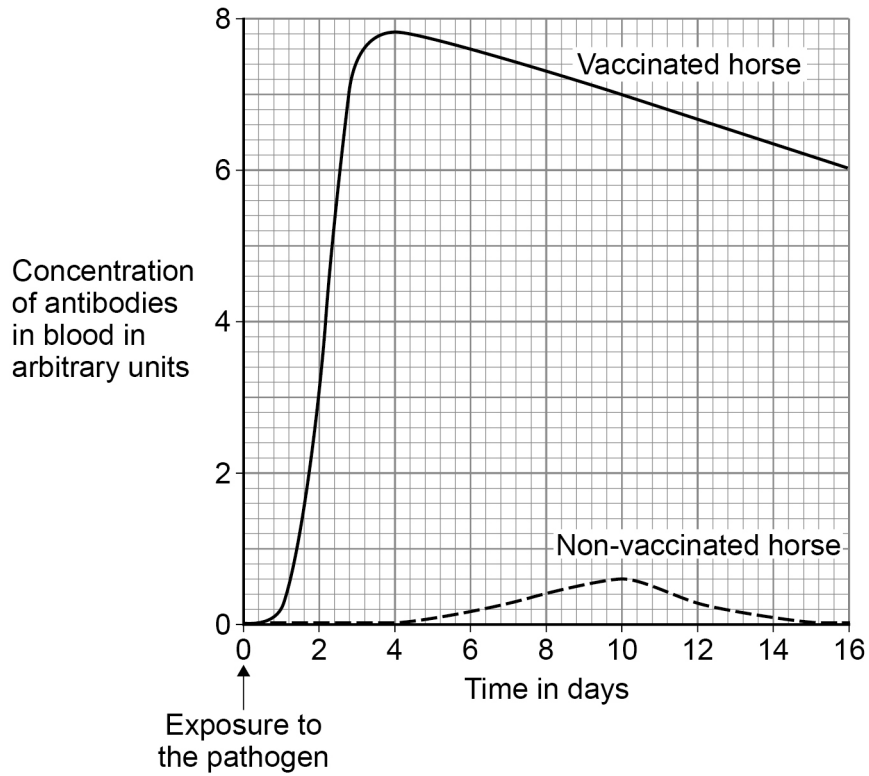
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0 4 . 6

Figure 9 shows the immune response of a vaccinated horse and the immune response of a non-vaccinated horse after exposure to the horse flu pathogen.

Figure 9



Describe the differences between the immune response of the vaccinated horse and the immune response of the non-vaccinated horse.

You should include data from **Figure 9**.

[4 marks]



0 5

People with diabetes may develop foot ulcers.

Foot ulcers are open sores that will not heal.

0 5 . 1

People with diabetes can have a blood glucose concentration which is too high.

Which hormone causes a high blood glucose concentration to decrease?

[1 mark]

0 5 . 2

There is a 24.7 % increased risk of developing foot ulcers for every 1 °C difference in the skin temperature of one foot compared to the other foot.

The skin temperatures of the feet of one person with diabetes are shown in **Table 4**.**Table 4**

	Left foot	Right foot
Skin temperature in °C	27.42	25.28

Calculate the increased risk of foot ulcers for the person in **Table 4**.**[3 marks]**

Increased risk = _____ %

Question 5 continues on the next page**Turn over ►**

0 5 . 3

An investigation was carried out for 6 months on 100 patients with diabetes.

They were split into two equivalent groups of fifty patients.

Table 5 shows the care plan for each group.

Table 5

Care plan for group 1	Care plan for group 2
• Every 12 weeks a nurse visually checks the feet. • The nurse uses the back of the hand to check the skin temperature of both feet. • The nurse provides preventative treatment if any inflammation or temperature difference is noted.	• The patient measures their skin temperature on the same six sites on each foot daily. • The patient uses a handheld infrared thermometer with an accuracy of $\pm 1.5^{\circ}\text{C}$. • The patient calculates the mean skin temperature for each foot. • The patient contacts a nurse if inflammation or a temperature difference of more than 2.2°C is recorded. • The nurse provides preventative treatment.

Inflammation in the feet can lead to foot ulcers.

Inflammation was recorded when a red swollen area was seen on either foot.

Results from the study showed:

- 20% of **Group 1** developed foot inflammation with 7 people getting an ulcer.
- 2% of **Group 2** developed foot inflammation with 1 person getting an ulcer.

The person who developed a foot ulcer in **Group 2** did not contact the nurse when a temperature difference of more than 2.2°C was recorded.



[6 marks]

[illegible]

10

Turn over ►



0 6

There are two tissues which transport substances around a plant.

Table 6 shows some information about the two tissues.

Table 6

	Tissue A	Tissue B
Structure	• Dead cells with no cytoplasm. • Cells have no end walls. • Cells have side walls thickened with a substance called lignin.	• Living cells with cytoplasm. • Cells have end walls with tiny holes called pores. • Cells have close contact with many mitochondria.
Function	To move water and mineral ions from the roots to the leaves.	To move dissolved sugars from the leaves to the rest of the plant.

0 6**. 1**

Name tissue **A** and tissue **B**.

[2 marks]

Tissue **A** _____

Tissue **B** _____

0 6**. 2**

Describe how the structure of tissue **A** and the structure of tissue **B** are related to their functions.

[4 marks]

Tissue **A** _____

Tissue **B** _____



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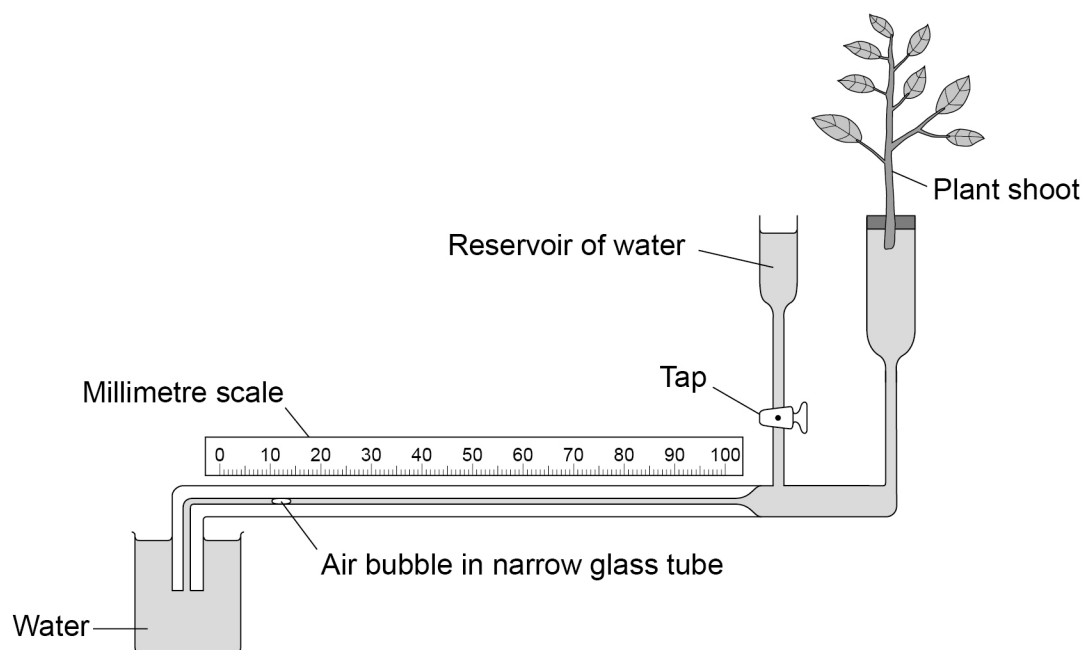
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One student used a potometer to investigate the rate of water uptake in a plant shoot.

Figure 10 shows a potometer.

Figure 10



As the plant shoot takes in water the air bubble moves.

The student measured the distance the air bubble had moved every two hours.

Table 7 shows the results.

Table 7

Time of day	Position of air bubble in mm
08:00	2
10:00	16
12:00	36
14:00	69
16:00	78
18:00	87
20:00	96
22:00	96

0 6 . 3

The rate of movement of the air bubble is directly related to the rate of water uptake by the plant shoot.

Calculate the mean rate of water uptake during the investigation.

Give the unit.

[3 marks]

Mean rate of water uptake = _____

Unit = _____

0 6 . 4

Another student did the same investigation at the same time but with a different shoot.

The air bubble moved at a slower rate.

Suggest why the air bubble moved at a slower rate with the different shoot.

[2 marks]

Question 6 continues on the next page

Turn over ►

Table 7 is repeated below.

Table 7

Time of day	Position of air bubble in mm
08:00	2
10:00	16
12:00	36
14:00	69
16:00	78
18:00	87
20:00	96
22:00	96

0 6 . 5 Describe the changes in the **rate** of water uptake during the investigation.

Use data from **Table 7**.

[3 marks]



0	6	.	6
---	---	---	---

Explain the change in the **rate** of water uptake from 08:00 to 12:00.

Daylight hours were from 06:30 to 19:00.

[4 marks]

18

Turn over for the next question

Turn over ►



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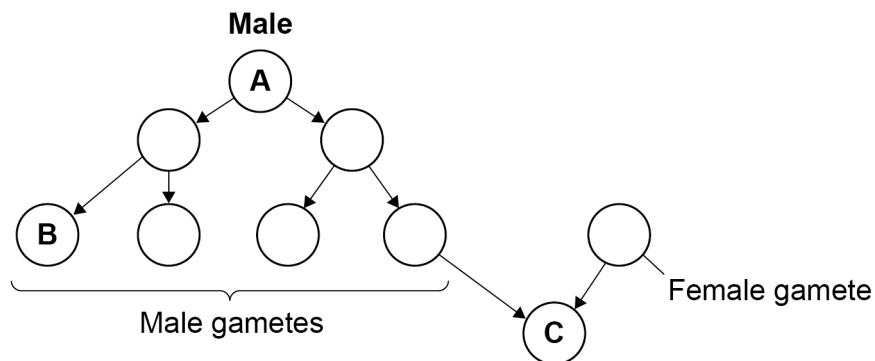
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0 7

Figure 11 shows fertilisation in an animal.

Figure 11



Male gametes are formed from cell A.

One of the male gametes fertilised a female gamete to produce cell C.

0 7 . 1

Cell A has 38 chromosomes.

Give the number of chromosomes in cell B and in cell C.

[2 marks]

B _____

C _____

0 7 . 2

Name the type of cell division which forms gametes.

[1 mark]

0 7 . 3

Describe cell division to form gametes.

[2 marks]

Turn over ►

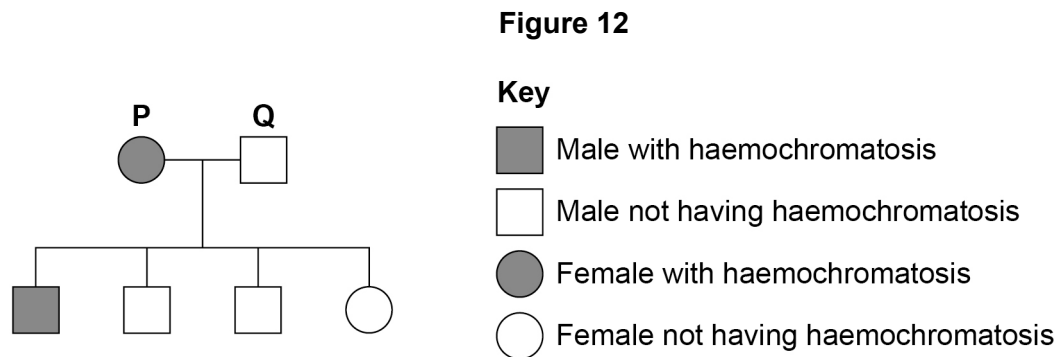


0 7 . 4

Haemochromatosis is an inherited disorder in humans.

The allele for haemochromatosis is recessive.

Figure 12 shows the inheritance of haemochromatosis in a family.



Person **P** and person **Q** plan to have another child.

Draw a genetic diagram to show the possible phenotypes of the child.

You should:

- identify the phenotypes of each offspring genotype
- use the following symbols:
H = allele for **not** having haemochromatosis
h = allele for having haemochromatosis.

[4 marks]



07.5

In **another** family the female parent does not have haemochromatosis and the male parent does have haemochromatosis.

The parents have had genetic testing and have been told that their children **cannot** have haemochromatosis

Explain why.

[3 marks]

12

END OF QUESTIONS



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